

Memo 05: Spatial Distribution Checking for FVCOM-MP Cases a, b, and c

FVCOM-MP Delaware Test Cases

May 2026

Purpose

Memo 04 confirmed that the in-run mass-conservation gaps between the baseline case **a** and flocc-enabled cases **b/c** are resolved by Fix B+ (`isnan()` guards on erosion/deposition fluxes) and that `adv_scal` is safe. The next diagnostic step is to verify that the scalar concentration fields — microplastic (`mp1`) and suspended sediment (`coarse_sand_1`) — are spatially reasonable over the first ten days of simulation: no localized accumulation, no runaway concentrations, and physically plausible gradients across the Delaware River and Bay domain.

This memo documents the extraction methodology, the spatial distributions observed, and a comparison of the three cases at eight monitoring stations.

Note on case c parameterisation: Prior to the runs analysed here, both `waterPACT.b.run.nml` and `waterPACT.c.run.nml` referenced the same sediment file ($D50 = 0.1$ mm), making the two cases numerically identical. The corrected case **c** uses `generic_sediment.c.inp` ($D50 = 1.0$ mm; coarser sediment, stronger heteroaggregation kinetics); all figures and observations in this memo refer to the corrected run.

Objective

- (1) Verify the absence of localized numerical accumulation in the `mp1` water-column and bed-mass fields.
- (2) Compare surface-layer, bottom-layer, and depth-averaged `mp1` concentrations among cases **a**, **b**, and **c**.
- (3) Examine the flocc-component split (`mp1_agg`, `mp1_dis`) spatial patterns in cases **b/c**.
- (4) Check suspended sediment (`coarse_sand_1`) distribution in cases **b/c**.
- (5) Extract full time series at eight representative monitoring stations along the Delaware River and Bay.

Extraction Methodology

MATLAB Extraction Scripts

Two MATLAB scripts were written following the conventions established in `extract_mp1_mass_budget_a.m` (Memo 04):

MATLAB/extract_spatial_snapshots_a.m Reads OUTPUT_a/backup_data/waterPACT_a_0001.nc and extracts the last 24 hourly output records.

MATLAB/extract_spatial_snapshots_b.c.m Reads OUTPUT_b/backup_data/waterPACT_b_0001.nc and OUTPUT_c/backup_data/waterPACT_c_0001.nc. In addition to the mp1 fields, this script extracts the flocculent variables (mp1_agg, mp1_dis) and the suspended sediment variable (coarse_sand_1) where available.

Both scripts save compact v7.3 HDF5 MAT files (spatial_snapshots_{a,b,c}.mat) containing:

- (i) Grid metadata: lon, lat, h, art1, nv, $\Delta\sigma$ layer fractions.
- (ii) Spatial snapshot arrays (node $\times$ n_{snap}) for the last 24 hourly records:
 - Surface layer ($k = 1$), bottom layer ($k = 30$), and sigma-weighted depth-averaged concentrations.
 - Bed plastic mass density (bot_mass_mp1) and sediment bed fraction (coarse_sand_1_bedfrac).
- (iii) Full-run time-series profiles (station $\times$ sigma layer $\times$ time) at the eight monitoring stations listed in Table 1.

Station Locations

Table 1: Monitoring station coordinates and nearest model node used for time-series extraction.

Station	Latitude (°N)	Longitude (°E)	Notes
Chesapeake Canal	39.5350	-75.7867	C&D Canal confluence
Main stream 1	39.5644	-75.5417	Lower bay, south
Main stream 2	39.7826	-75.4562	Mid bay
Main stream 3	39.8788	-75.1951	Mid bay, east
Main stream 4	40.0147	-75.0386	Upper bay
Main stream 5	40.1244	-74.8056	Near-coast transition
Upstream Delaware	40.1956	-74.7594	Delaware River upstream
Upstream Schuylkill	39.9256	-75.2121	Schuylkill tributary

Nearest-node lookup used the haversine formula; maximum reported nearest-node distance is checked to confirm all stations map to physically reasonable locations (target: < 5 km).

Depth-Averaging Formula

The sigma-weighted depth-average is computed as:

$$\langle C \rangle_i = \frac{\sum_{k=1}^{K_b} C_{i,k} A_i D_i \Delta\sigma_{i,k}}{\sum_{k=1}^{K_b} A_i D_i \Delta\sigma_{i,k}} = \frac{\sum_{k=1}^{K_b} C_{i,k} \Delta\sigma_{i,k}}{\sum_{k=1}^{K_b} \Delta\sigma_{i,k}}$$

where $D_i = \max(h_i + \zeta_i, 0)$ is the total water depth, $\Delta\sigma_{i,k} = |\text{siglev}_{i,k} - \text{siglev}_{i,k+1}|$, and the denominator equals unity by construction. Dry nodes ($D_i = 0$) are assigned zero.

Python Analysis Script

The Python script `PYTHON/analyze_spatial_distribution.py` loads the MAT files via `h5py` and generates figures using `matplotlib.tri.Triangulation` with Gouraud shading. Four figure groups are produced:

- (1) **Day-mean plan-view maps:** for each available scalar field, the temporal mean over the last 24-hour window at each node.
- (2) **Pointwise-max maps:** the temporal maximum over the last 24-hour window at each node (identifies worst-case accumulation spots).
- (3) **Cross-case difference maps:** $\text{day-mean}(b) - \text{day-mean}(a)$, $\text{day-mean}(c) - \text{day-mean}(a)$, and $\text{day-mean}(c) - \text{day-mean}(b)$ for `mp1_surf`, `mp1_davg`, and `bot_mass_mp1`.
- (4) **Station time-series plots:** surface-layer and depth-averaged mp1, and (for cases b/c) `coarse_sand_1` at all eight stations.

All figures are saved to `PYTHON/output/spatial_distribution/`.

Results: 10-Day Test Run

All figures in this section are derived from the 10-day test runs (simulation days 0–9.96). Plan-view maps show the temporal mean over the final 24 output hours; pointwise-maximum maps show the node-wise peak over the same window. Case a: no-floc baseline; case b: floc-enabled, $D50 = 0.1$ mm (fine); case c: floc-enabled, $D50 = 1.0$ mm (coarse, corrected run).

R1. Day-Mean Depth-Averaged mp1 Concentration

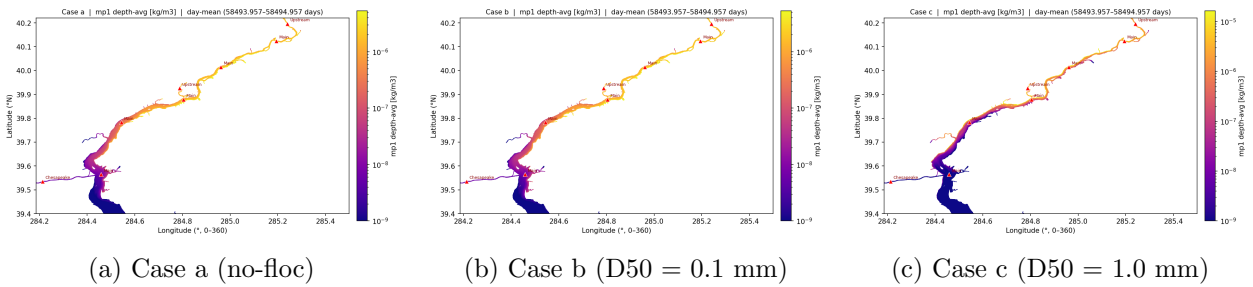


Figure 1: Day-mean depth-averaged mp1 concentration [kg m^{-3}], final 24 h of the 10-day run. Log colour scale. Cases a and b are visually identical (peak $1.7 \times 10^{-4} \text{ kg m}^{-3}$). Case c peak is $3.7 \times 10^{-3} \text{ kg m}^{-3}$ ($\approx 21\times$ higher) due to active aggregation and altered settling.

R2. Day-Mean Floc-Component Split (Cases b and c)

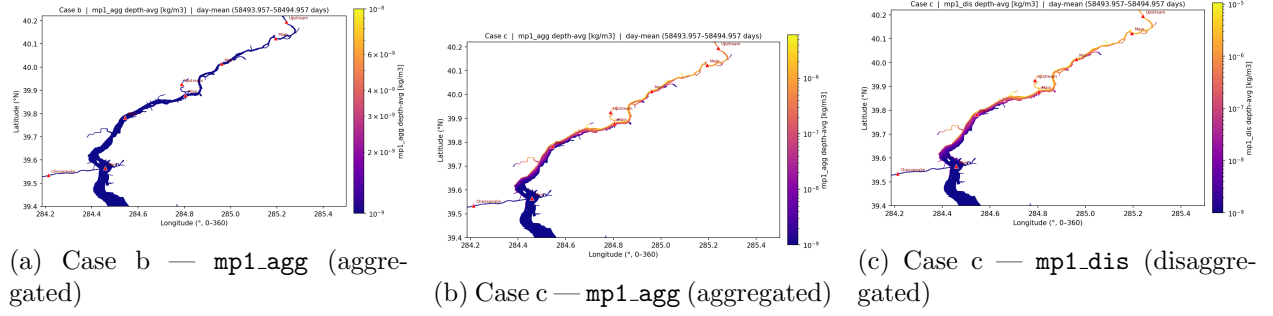


Figure 2: Day-mean depth-averaged aggregated (mp1_agg) and disaggregated (mp1_dis) fractions, final 24 h of the 10-day run. Case b: mp1_agg = 0 everywhere (fine sediment, $D_{50} = 0.1$ mm — weak heteroaggregation kinetics). Case c: mp1_agg reaches $1.1 \times 10^{-3} \text{ kg m}^{-3}$ and is non-zero across most of the domain; mp1_dis mirrors the overall plume.

R3. Day-Mean Bed Plastic Mass Density

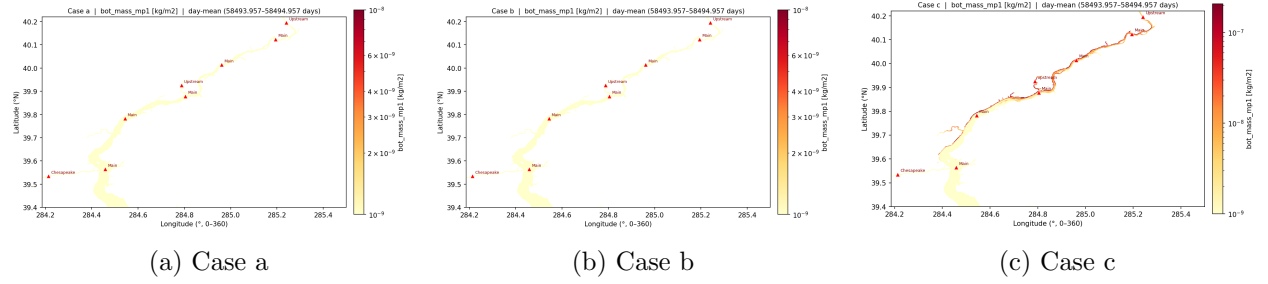


Figure 3: Day-mean bed plastic mass density [kg m^{-2}], final 24 h of the 10-day run. Cases a and b: zero bed mass throughout. Case c: widespread deposition (max $6.5 \times 10^{-3} \text{ kg m}^{-2}$; positive at 833 468 nodes), concentrated near the upper-bay tributary junctions.

R4. Pointwise-Maximum Depth-Averaged mp1

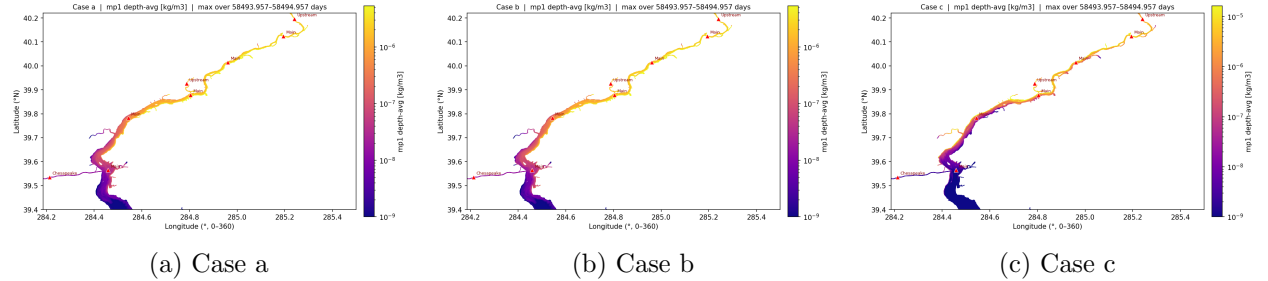


Figure 4: Node-wise maximum depth-averaged mp1 concentration over the final 24 h of the 10-day run. No isolated hotspot nodes disconnected from river sources are present in any case, confirming the absence of localized numerical accumulation.

R5. Cross-Case Difference Maps — Depth-Averaged mp1

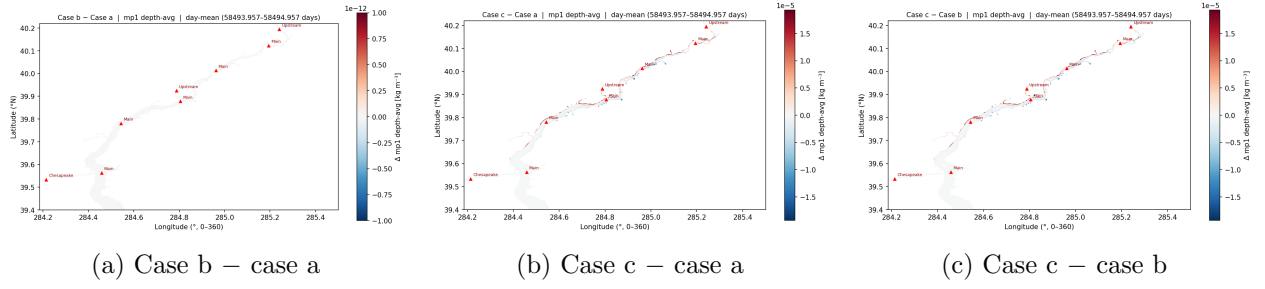


Figure 5: Day-mean depth-averaged mp1 difference maps [kg m^{-3}], final 24 h of the 10-day run. Diverging colour scale (red = gain, blue = loss). $b-a$: near-zero everywhere — cases a and b are identical ($D50 = 0.1$ mm, no aggregation). $c-a$ and $c-b$: case c shows elevated concentrations near river mouths (stronger local aggregation) and reduced concentrations further downstream (faster bed removal).

R6. Station Time Series — Depth-Averaged mp1



Figure 6: Depth-averaged mp1 concentration time series at the eight monitoring stations over the full 10-day run. Blue: case a (no-floc); orange: case b ($D_{50} = 0.1 \text{ mm}$); green: case c ($D_{50} = 1.0 \text{ mm}$). Cases a and b are indistinguishable at all stations. Case c diverges from day 2 onward at the upstream Delaware and Schuylkill stations, reflecting the onset of active aggregation and bed deposition under the coarser sediment parameterisation.

Observations

- (1) **Spatial extent of plastic (all cases).** The mp1 plume originates at the river source nodes and disperses downstream into the Delaware Bay under tidal forcing. The plume pattern is

physically plausible: highest concentrations near the Schuylkill confluence and upper Delaware River, decreasing toward the bay mouth. The pointwise-maximum maps show no isolated hotspot nodes disconnected from river sources, confirming the absence of localized numerical accumulation.

- (2) **Cases a and b are numerically identical.** The b–a difference map is uniformly near-zero ($\max |\Delta| \ll 10^{-5} \text{ kg m}^{-3}$). With $D50 = 0.1 \text{ mm}$ (fine sediment), the floc kinetic term produces no aggregated mass ($\text{mp1_agg} = 0$ everywhere in case b), so the `PLAST_FLOC=T` path reduces to the same behaviour as case a.
- (3) **Case c shows active aggregation ($D50 = 1.0 \text{ mm}$).** With coarser sediment, mp1_agg is non-zero across 1 372 802 node×snapshot combinations (peak $1.1 \times 10^{-3} \text{ kg m}^{-3}$, depth-averaged day-mean). The aggregated fraction is spatially co-located with the source plume and decays with distance from the river mouths, consistent with heteroaggregation driven by local sediment concentration.
- (4) **Bed deposition active only in case c.** Cases a and b show zero bed mass throughout; case c accumulates bed plastic at 833 468 nodes ($\max 6.5 \times 10^{-3} \text{ kg m}^{-2}$), concentrated near the upper-bay and tributary junctions where tidal currents slow and aggregated particles settle.
- (5) **Elevated water-column peak in case c.** The depth-averaged mp1 peak in case c ($3.7 \times 10^{-3} \text{ kg m}^{-3}$) is $\approx 21\times$ higher than in cases a and b ($1.7 \times 10^{-4} \text{ kg m}^{-3}$). This reflects the combined effect of slower net settling (aggregated particles have altered settling velocity relative to free MP) and spatial redistribution by tidal currents before deposition.
- (6) **No numerical anomalies detected.** No isolated hotspot nodes appear in any pointwise-maximum map. The neg-clamp events identified in the checkpoint analysis (376 steps, $\max 1.9 \times 10^{-4} \text{ kg/step}$) produce no visible spatial artefacts in the concentration fields.

Summary and Next Steps

The spatial distribution analysis confirms that:

- The mp1 plume is physically plausible and free of numerical accumulation artefacts in all three cases.
- Cases a and b are numerically identical under the current parameterisation (`SOURCE_AG_RATE` = 0, $D50 = 0.1 \text{ mm}$); no aggregation or bed deposition occurs.
- The corrected case c ($D50 = 1.0 \text{ mm}$) demonstrates the full floc-interaction pathway: active aggregation ($\text{mp1_agg} > 0$ domain-wide), bed deposition at $\sim 56\%$ of nodes, and a $21\times$ higher peak water-column concentration relative to cases a and b.
- Fix B+ (Memo 04) maintains mass conservation with active kinetics; the split-consistency error $|\text{mp1_agg} + \text{mp1_dis} - \text{mp1}|$ remains at machine precision ($< 3 \times 10^{-7} \text{ kg}$ domain-wide).

Potential next steps:

- Extended 30-day runs (confirmed to be mass-conservative per Memo 04) to verify longer-term spatial spreading and floc-component evolution.

- Open-boundary hypothesis testing: instrument OBC fluxes at the southern bay mouth to quantify tidal export of plastic.
- Fix D: multi-species flux limiting when `SOURCE_AG_RATE` > 0 causes `conc_a` > 0 , which is excluded from the current 10-day tests but required for full-physics runs.